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AI-DRIVEN

CYBER SECURITY

YOUR AI APPLICATION MIGHT BE PERFECTLY CODED, YET STILL BE VULNERABLE BECAUSE OF THE MODEL YOU DOWNLOADED, THE DATASET USED TO TRAIN IT, OR A THIRD-PARTY LIBRARY YOU INSTALLED.

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WHAT IS AI SUPPLY CHAIN SECURITY?

- PROTECTS EVERY COMPONENT USED TO BUILD, TRAIN, DEPLOY, AND OPERATE AN AI SYSTEM.
- INCLUDES MODELS, DATASETS, LIBRARIES, APIS, PLUGINS, AND EXTERNAL SERVICES.
- GOAL: PREVENT COMPROMISED, MALICIOUS, OR UNTRUSTED COMPONENTS FROM ENTERING AN AI APPLICATION.

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BASE MODEL CHECKPOINT VALIDATION

- A MODEL CHECKPOINT CONTAINS THE MODEL'S LEARNED PARAMETERS, COMMONLY CALLED WEIGHTS.
- VERIFY THE MODEL'S PUBLISHER, OFFICIAL SOURCE, VERSION, AND INTEGRITY BEFORE DOWNLOADING OR DEPLOYING IT.
- USE CRYPTOGRAPHIC HASHES OR TRUSTED SIGNATURES TO DETECT UNEXPECTED CHANGES.
- PREFER SAFE MODEL FORMATS AND LOADING METHODS.

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HUGGING FACE MODEL AND DATASET LINEAGE

- MODEL LINEAGE TRACKS WHERE A MODEL CAME FROM, HOW IT WAS TRAINED, AND WHICH VERSIONS OR DATASETS IT DEPENDS ON.
- CHECK THE MODEL CARD, DATASET SOURCES, LICENSE, COMMIT HISTORY, AND DEPENDENCIES.
- PIN A SPECIFIC REVISION INSTEAD OF AUTOMATICALLY TRUSTING THE LATEST VERSION.
- REMEMBER: A MODEL CARD OR POPULAR DOWNLOAD COUNT DOES NOT GUARANTEE SAFETY.

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DEPENDENCY AND INTEGRATION SECURITY

- AI SYSTEMS OFTEN RELY ON PYTHON PACKAGES, APIS, PLUGINS, AND EXTERNAL SERVICES.
- A COMPROMISED OR VULNERABLE DEPENDENCY CAN PUT THE ENTIRE APPLICATION AT RISK.
- PIN PACKAGE VERSIONS, SCAN DEPENDENCIES, PROTECT API KEYS, AND REVIEW PERMISSIONS.
- KEEP DEPENDENCIES UPDATED THROUGH A CONTROLLED PROCESS.

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COMMON SUPPLY CHAIN THREATS

- MALICIOUS OR TAMPERED MODEL WEIGHTS.
- POISONED OR UNTRUSTED TRAINING DATASETS.
- VULNERABLE OR COMPROMISED LIBRARIES.
- UNTRUSTED MODEL-LOADING CODE AND THIRD-PARTY INTEGRATIONS.
- LICENSE VIOLATIONS AND EXPOSED CREDENTIALS.

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WHY AI SECURITY ENGINEERS NEED IT

- IDENTIFY RISKS BEFORE MODELS ENTER PRODUCTION.
- VERIFY PROVENANCE, INTEGRITY, AND DEPENDENCIES.
- MAINTAIN MODEL AND DATASET INVENTORIES.
- MONITOR CHANGES AND INVESTIGATE SUSPICIOUS COMPONENTS.
- CREATE A REPEATABLE APPROVAL PROCESS BEFORE DEPLOYMENT.